Lecture 16

AGENDA

- Conditionally randomized experiments with discrete blocking variables
- Conditionally randomized experiments/observational studies with continuous blocking/control variables
- Omitted variable bias revisited

Conditionally Randomized Experiments with a Discrete Blocking Variable

Conditionally Randomized Experiment

Suppose for one group/block/stratum ($X = 1$) a coin was flipped to assign treatment to half of the units, and for another group/block stratum ($X = 0$), one out of four units was assigned to treatment. This is called a blocked, stratified, or conditionally randomized experiment. Essentially, we now have two experiments.

i	X_i	D_i	Y_i	$Y_i(1)$	$Y_i(0)$	$Y_i(1) - Y_i(0)$
1	0	1	70	70	?	?
2	1	1	80	80	?	?
3	0	0	60	?	60	?
4	1	0	70	?	70	?
5	0	0	70	?	70	?
6	0	0	80	?	80	?

Conditionally Randomized Experiment

Within each of our two groups ($X = 0$ and $X = 1$), all the properties discussed in for randomized D hold.

i	X_i	D_i	Y_i	$Y_i(1)$	$Y_i(0)$	$Y_i(1) - Y_i(0)$
1	0	1	70	70	70	0
2	1	1	80	80	70	10
3	0	0	60	60	60	0
4	1	0	70	80	70	10
5	0	0	70	70	70	0
6	0	0	80	80	80	0

remember red numbers not observed

What is the average $Y_i(1)$ among the $X = 0$ subjects?

Conditionally Randomized Experiment

Within each of our two groups ($X = 0$ and $X = 1$), all the properties discussed in for randomized D hold.

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3	0	0	60	60	60	0
4	1	0	70	80	70	10
5	0	0	70	70	70	0
6	0	0	80	80	80	0

remember red numbers not observed

What is the average $Y_i(1)$ among the $X = 0$ subjects?

$$\frac{70 + 60 + 70 + 80}{4} = 70$$

Conditionally Randomized Experiment

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6	0	0	80	80	80	0

remember red numbers not observed

What is the average $Y_i(1)$ among the $X = 0$, $D = 1$ subjects?

Conditionally Randomized Experiment

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What is the average $Y_i(1)$ among the $X = 0$, $D = 1$ subjects?

$$\frac{70}{1} = 70$$

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4	1	0	70	80	70	10
5	0	0	70	70	70	0
6	0	0	80	80	80	0

remember red numbers not observed

What is the average $Y_i(0)$ among the $X = 0, D = 0$ subjects?

$$\frac{60 + 70 + 80}{3} = 70$$

CONDITIONAL ATE FOR $X = 0$

Because we have a randomized experiment within the $X = 0$ group and because of the linearity of expectations, the SLR slope (difference in means) among the $X = 0$ group provides an unbiased estimator of the conditional average treatment effect ($CATE_{x=0}$).

$$\begin{aligned}
&\frac{70}{1} - \frac{60 + 70 + 80}{3} \\
&= \frac{70 + 60 + 70 + 80}{4} - \frac{70 + 60 + 70 + 80}{4} \\
&= \frac{0 + 0 + 0 + 0}{4}
\end{aligned}$$

Conditionally Randomized Experiment

Within the $X = 1$ group, all the properties discussed for randomized D hold.

i	X_i	D_i	Y_i	$Y_i(1)$	$Y_i(0)$	$Y_i(1) - Y_i(0)$
1	0	1	70	70	70	0
2	1	1	80	80	70	10
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What is the average $Y_i(1)$ among the $X = 1$ subjects?

$$\frac{80 + 80}{2} = 80$$

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4	1	0	70	80	70	10
5	0	0	70	70	70	0
6	0	0	80	80	80	0

What is the average $Y_i(1)$ among the $X = 1, D = 1$ subjects?

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What is the average $Y_i(1)$ among the $X = 1$, $D = 1$ subjects?

$$\frac{80}{1} = 80$$

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$$
\begin{aligned}
&\frac{80}{1} - \frac{70}{1} \\
&= \frac{80 + 80}{2} - \frac{70 + 70}{2} \\
&= \frac{10 + 10}{2}
\end{aligned}
$$

Standardization via APD

If we average the $CATE_{x=0}$ and $CATE_{x=1}$ by the proportion in each group/block/stratum, this is an APD, which gives us the ATE. In epidemiology, this is known as standardization.

i	X_i	D_i	Y_i	$Y_i(1)$	$Y_i(0)$	$Y_i(1) - Y_i(0)$
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$$\frac{4}{6} \cdot \left\{ \frac{70}{1} - \frac{60+70+80}{3} \right\} + \frac{2}{6} \cdot \left\{ \frac{80}{1} - \frac{70}{1} \right\}$$
$$= \frac{20}{6}$$

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6	0	0	80	80	80	0

$$\frac{4}{6} \cdot \left\{ \frac{70}{1} - \frac{60+70+80}{3} \right\} + \frac{2}{6} \cdot \left\{ \frac{80}{1} - \frac{70}{1} \right\}$$

$$= \frac{20}{6} = \frac{10+10}{6}$$

- The ATE estimation method on the previous slide is equivalent to an APD with an interactive linear model ($Y \sim D + X + D : X$ in R formula notation).
- Many researchers would simply estimate the effect with an additive linear model ($Y \sim D + X$ in R formula notation), but this would target a weighted average treatment effect.

Omitted variable bias with SLR

If we ignore the groups, and run an SLR on the entire data, we get omitted variable bias.

i	X_i	T_i	Y_i	$Y_i(1)$	$Y_i(0)$	$Y_i(1) - Y_i(0)$
1	0	1	70	70	70	0
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What is the average $Y_i(1) - Y_i(0)$?

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3	0	0	60	60	60	0
4	1	0	70	80	70	10
5	0	0	70	70	70	0
6	0	0	80	80	80	0

What is the average $Y_i(1) - Y_i(0)$?

$$\begin{aligned}\frac{0+10+0+10+0+0}{6} &= \frac{20}{6} \\ &\approx 3.3 \\ &\neq \frac{70+80}{2} - \frac{60+70+70+80}{4}\end{aligned}$$

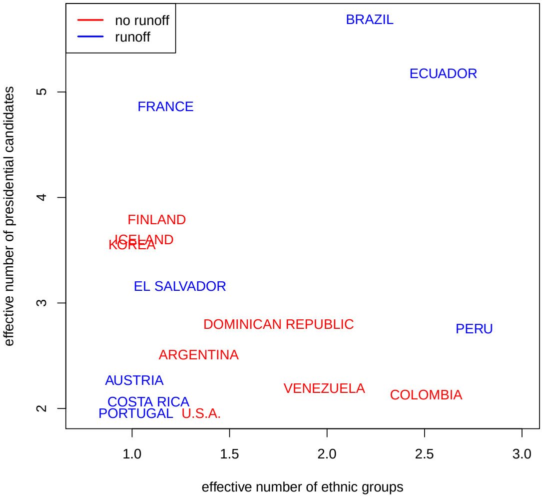
Summary of Conditionally Randomized Experiments

- A conditionally randomized experiment (with a discrete blocking variable) can be analyzed as multiple experiments and these results can be combined with standardization (an APD).
- A additive MLR of Y on D and X will produce a weighted average treatment effect.
- If you ignore the groups in a conditionally randomized experiment, and the probability of treatment assignment is different in the groups, then you will induce omitted variable bias.

Up Next

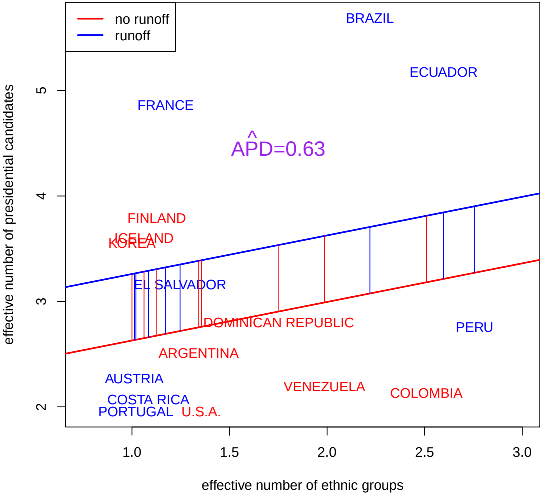
- What about randomization conditional on a continuous covariate and/or many covariates?
- Many non-experimental studies can be thought of as approximating a conditionally randomized experiment.
- We will use APDs to estimate the ATE in these studies (e.g. effect of runoff elections on number of presidential candidates).
- Need to worry about omitted variable bias.

Neto and Cox (1997) Table 3 Data

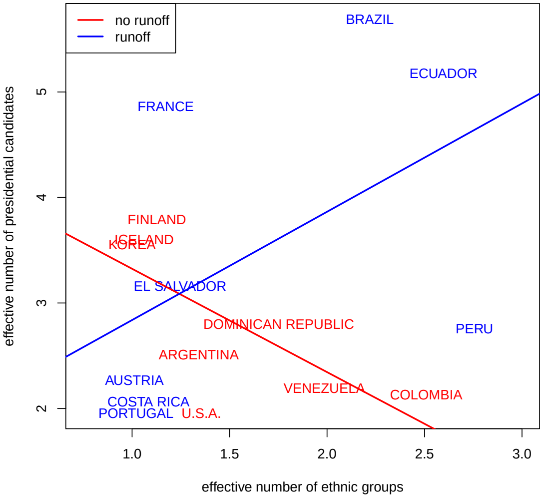


Remember the estimation of APDs for a binary covariate.

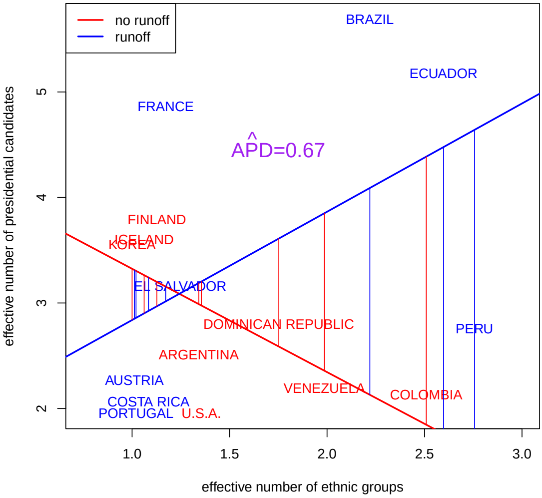
Average Partial Derivative (Runoff) in Additive Model



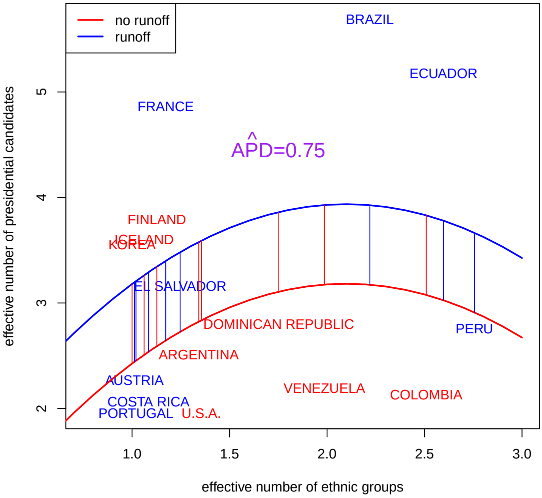
APD IN INTERACTIVE/UNPOOLED LINEAR MODEL



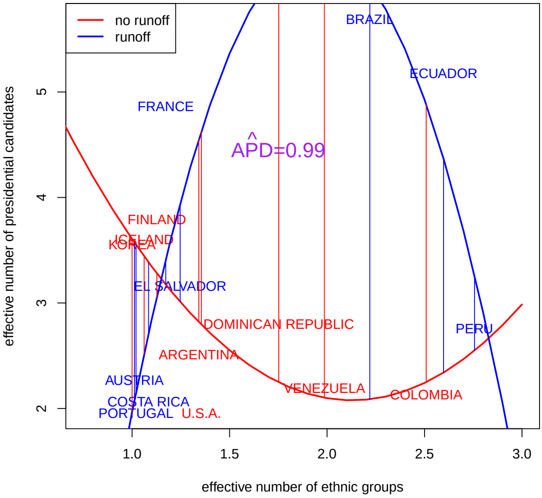
APD in Interactive/Unpooled Linear Model



APD in Additive Quadratic Models

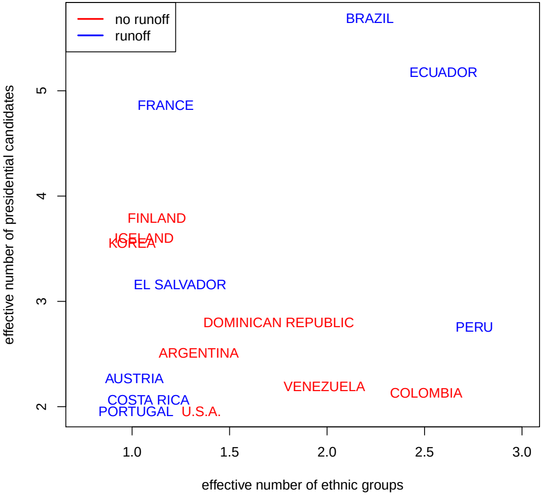


APD in Interactive/Unpooled Quadratic

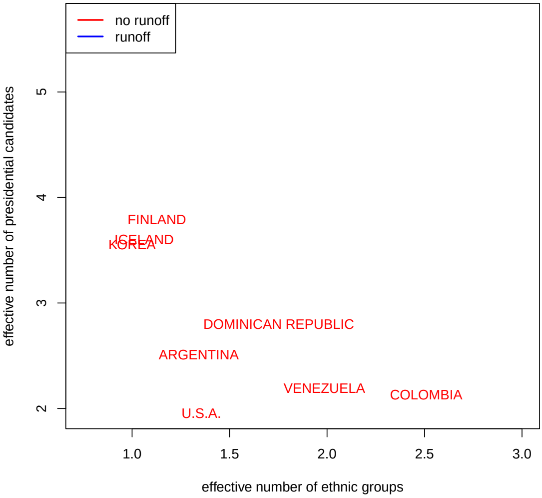


Let's think about the Neto and Cox data causally, and start with the heroic assumption that whether a country has a runoff election is randomly assigned, conditional on how many ethnic groups they have.

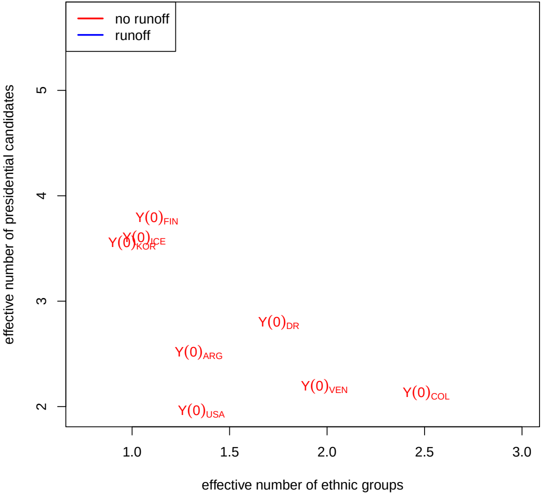
NETO AND COX (1997) TABLE 3 DATA



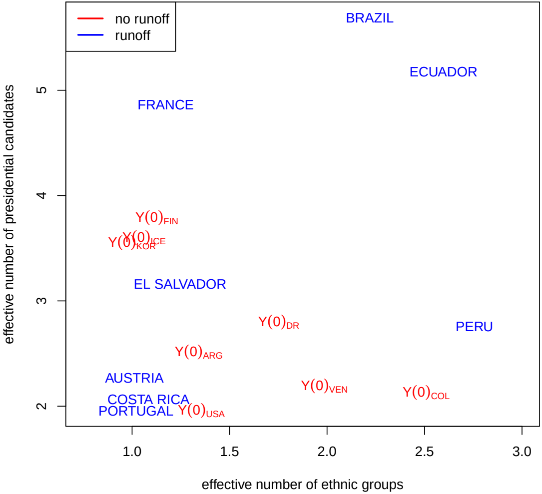
Focus on the no runoff countries



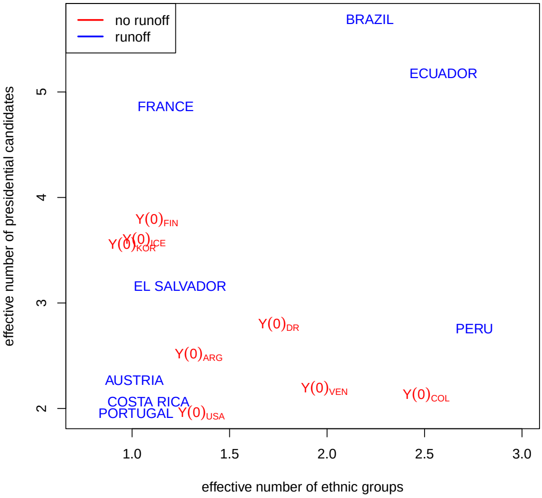
Which potential outcome are we observing?



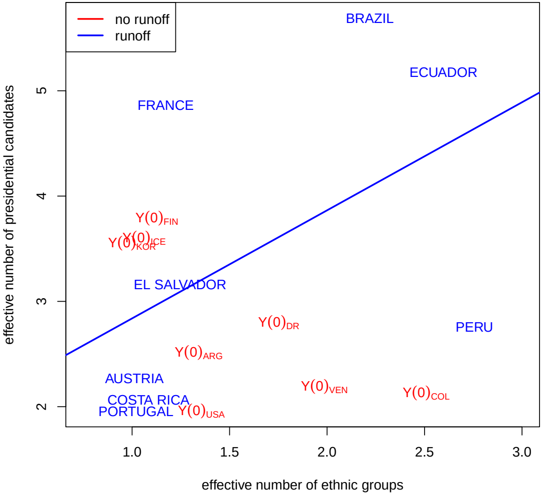
Bringing back runoff countries



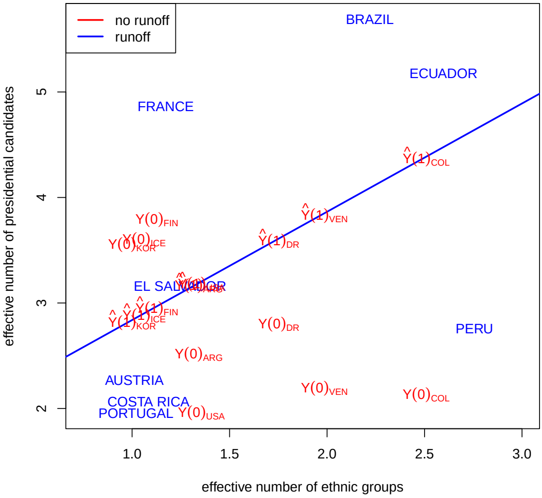
How to impute $Y(1)$ for no runoff countries?



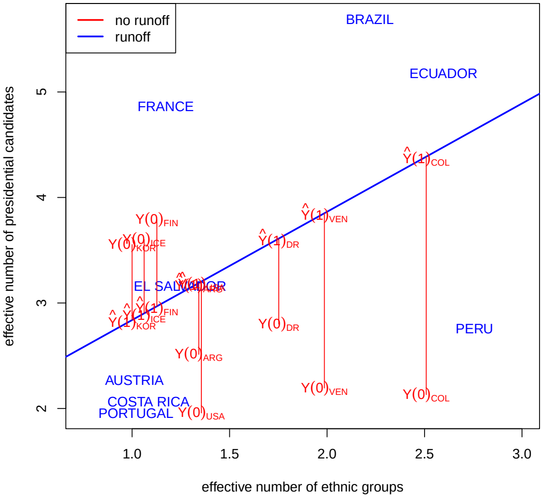
Here is one way (with linear regression)



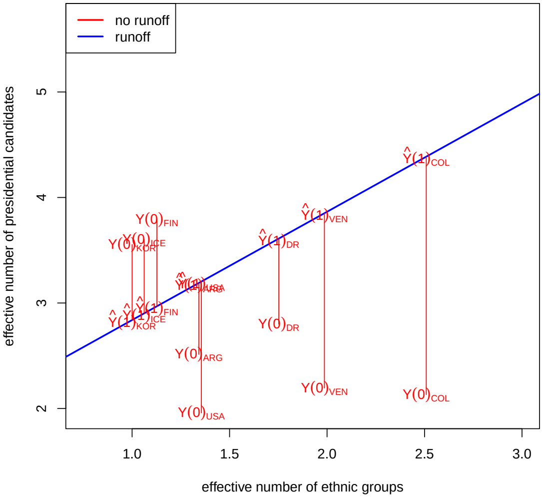
Imputations on the line



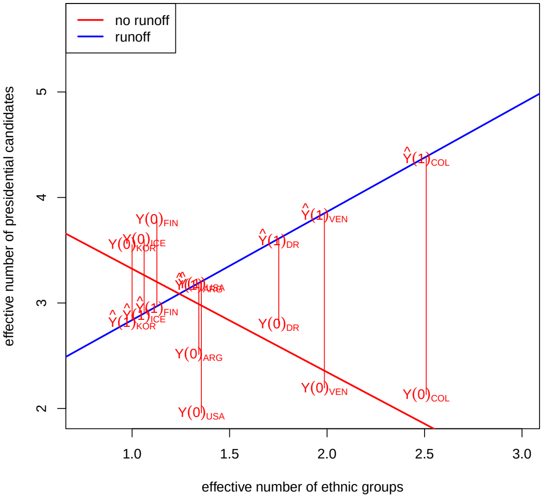
Imputed Individual Treatment Effects



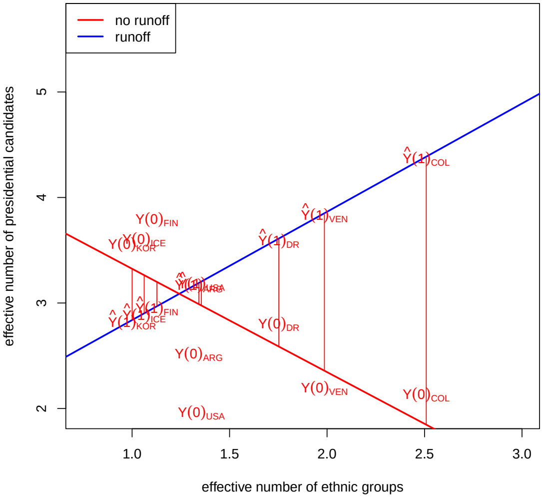
An important identity (part 1)



An important identity (part 2)

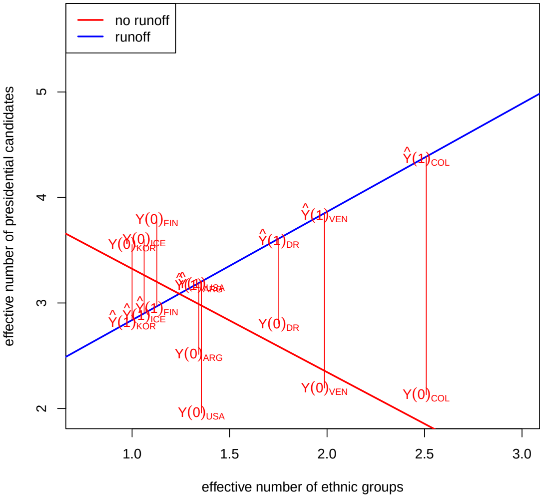


An important identity (part 3)

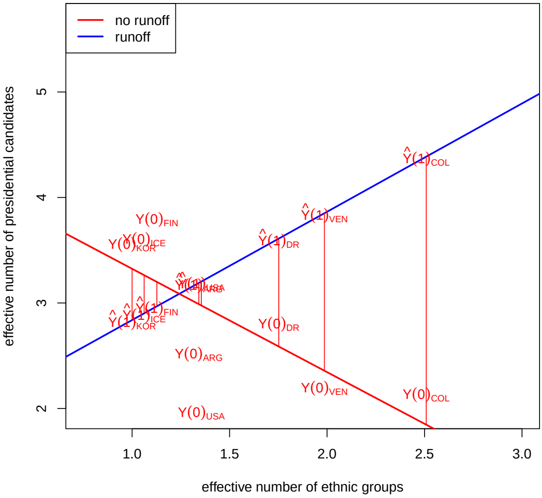


... because the residuals sum to zero.

$\widehat{ATE}$ AMONG NON RUNOFF COUNTRIES, EQUALS ...

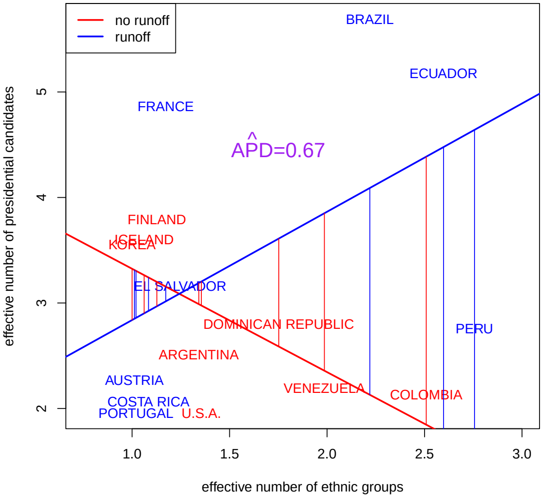


$\widehat{APD}$ AMONG NON RUNOFF COUNTRIES

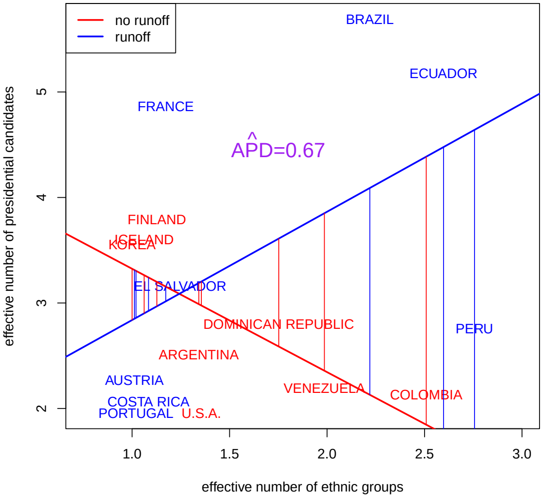


This works the same way for the runoff countries, so the $\widehat{APD}$ equals the $\widehat{ATE}$

$\widehat{APD}$ IN UNPOOLED LINEAR MODEL ...



… IS THE $\widehat{ATE}$ IN UNPOOLED LINEAR MODEL



Omitted Variable Bias

- we have been making the heroic assumption that whether a country has a runoff election is randomly assigned, conditional on how many ethnic groups they have.

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- concerns about 1 are essentially concerns about omitted variable bias

Recall this Omitted Variable Bias slide

Suppose we want B_1 from $y_i = B_0 + B_1 x_i + B_2 z_i + e_i$, but we run the regression of y on x alone. Suppose also that we have de-meaned x_i and z_i such that $\bar{x} = 0$ and $\bar{z} = 0$. Then,

$$\frac{\sum_{i=1}^{n} x_i y_i}{\sum_{i=1}^{n} x_i^2} =$$

$$=$$

$$=$$

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$$\frac{\sum_{i=1}^{n} x_i y_i}{\sum_{i=1}^{n} x_i^2} = \frac{\sum_{i=1}^{n} x_i (B_0 + B_1 x_i + B_2 z_i + e_i)}{\sum_{i=1}^{n} x_i^2}$$

$$=$$

$$=$$

Recall this Omitted Variable Bias slide

Suppose we want B_1 from $y_i = B_0 + B_1 x_i + B_2 z_i + e_i$, but we run the regression of y on x alone. Suppose also that we have de-meaned x_i and z_i such that $\bar{x} = 0$ and $\bar{z} = 0$. Then,

$$\frac{\sum_{i=1}^{n} x_i y_i}{\sum_{i=1}^{n} x_i^2} = \frac{\sum_{i=1}^{n} x_i (B_0 + B_1 x_i + B_2 z_i + e_i)}{\sum_{i=1}^{n} x_i^2}$$

$$= \frac{\sum_{i=1}^{n} x_i B_0}{\sum_{i=1}^{n} x_i^2} + \frac{\sum_{i=1}^{n} x_i B_1 x_i}{\sum_{i=1}^{n} x_i^2} + \frac{\sum_{i=1}^{n} x_i B_2 z_i}{\sum_{i=1}^{n} x_i^2} + \frac{\sum_{i=1}^{n} x_i e_i}{\sum_{i=1}^{n} x_i^2}$$

$$=$$

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$$\frac{\sum_{i=1}^{n} x_i y_i}{\sum_{i=1}^{n} x_i^2} = \frac{\sum_{i=1}^{n} x_i (B_0 + B_1 x_i + B_2 z_i + e_i)}{\sum_{i=1}^{n} x_i^2}$$

$$= \frac{\sum_{i=1}^{n} x_i B_0}{\sum_{i=1}^{n} x_i^2} + \frac{\sum_{i=1}^{n} x_i B_1 x_i}{\sum_{i=1}^{n} x_i^2} + \frac{\sum_{i=1}^{n} x_i B_2 z_i}{\sum_{i=1}^{n} x_i^2} + \frac{\sum_{i=1}^{n} x_i e_i}{\sum_{i=1}^{n} x_i^2}$$

$$= B_1 + B_2 \cdot \frac{\sum_{i=1}^{n} x_i z_i}{\sum_{i=1}^{n} x_i^2}$$

Potential Omitted Variables

- colonial history?
- ...

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The omitted variable bias formulas we presented earlier give a "back of the envelope" method for considering the implications of omitted variables (see QTM electives for more precise approaches).